Multi Scale Geometric Optimization of the Extrusion Die Angle

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April 20, 2026

Preface

This report documents the theoretical formulation, mathematical proofs, and computational implementation of the Multi-Scale Adaptive Geometric Strategy (MSAGS) for extrusion die optimization. Developed as part of the Manufacturing Technology (ManTech) curriculum at the Department of Mechanical Engineering, Indian Institute of Technology (IIT) Guwahati , this project seeks to bridge the gap between classical continuum mechanics and modern, accessible computational engineering.

The genesis of this project stemmed from a critical observation in bulk metal forming: traditional conical dies inherently force a rigid compromise between interfacial friction and internal shearing, imposing unavoidable energy penalties. Recognizing the severe consequences of this geometry—ranging from inflated power consumption to rapid tool wear and dead metal zone formation —we set out to mathematically prove and computationally model a superior alternative. By deriving a streamlined, third-order polynomial profile, we demonstrate how continuous boundary conditions can fundamentally bypass the physical limitations of linear die convergence.

The culmination of this research is the Extrusion Die Optimiser. Built entirely as a zero-latency, browser-based architecture, this application serves as an active decision-support system, empowering process engineers to rapidly evaluate die geometries, forecast thermal limits, and optimize extrusion parameters without the computational overhead of standard finite element simulations. Also you can find the link to the website here.

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April 20, 2026

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1 Introduction: The Critical Role of Die Angle

In the industrial manufacturing landscape, extrusion presses operate under immense pressure, often exerting thousands of tons of force. At the heart of this massive operation lies a surprisingly subtle geometric parameter: the extrusion die angle (α). Far from being a mere structural dimension, the die angle is the primary governor of stress distribution, metal flow characteristics, and the final mechanical integrity of the extruded product.

If this angle is not properly optimized, the consequences are severe. A non-optimal die angle can inflate energy consumption by 20% to 30% [7]. Furthermore, excessive force accelerates thermal fatigue and rapid die wear, leading to costly maintenance downtime. Geometrically, incorrect angles can also induce "dead metal zones"—areas where material stagnates within the die—resulting in surface defects, internal voids, and non-uniform grain structures [8].

1.1 The Physics of Deformation

To understand how to optimize the die angle, we must zoom in on the internal physics of the material as it is forced through the converging die. The total work performed during extrusion is essentially a summation of three energy components [3]:

- **Ideal Work (W_i):** The energy required for the homogeneous deformation of the material. This depends on the extrusion ratio (R) and the material's flow stress, but it remains independent of the die angle itself.
- **Friction Work (W_f):** The energy lost to friction at the billet-die interface.
- **Redundant Work (W_r):** The energy consumed by internal shear and velocity discontinuities as the material is forced to change its flow direction.

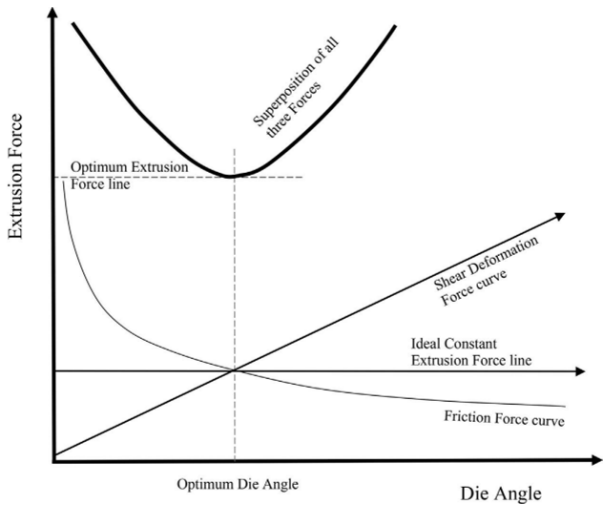


Figure 1: Die Angle VS Extrusion Force

Herein lies the core optimization problem—a classic trade-off. If we decrease the die angle to make the convergence gentler, we increase the surface contact area, which drives up the friction work (W_f). Conversely, if we increase the die angle to reduce the contact

area, the material undergoes a more abrupt change in flow direction, generating severe shearing strains and driving up the redundant work (W_r). The optimal die angle exists exactly at the point where the combined sum of friction and redundant work ($W_f + W_r$) is minimized.

1.2 Limitations of Historical and Current Approaches

Historically, the quest to determine the optimal extrusion die angle has relied heavily on classical analytical mechanics to establish a baseline energy balance.

Early approaches, such as the Slab Method [3], provided a straightforward force balance based on the assumption of homogeneous deformation. However, this method exhibits a critical shortfall: it completely disregards redundant work, or internal shearing. By failing to account for the energy consumed by velocity discontinuities, the Slab Method becomes highly inaccurate for evaluating larger die angles. To address this, Avitzur's application of the Upper Bound Theorem [11, 2] established a kinematically admissible velocity field, deriving a mathematical approximation for the optimum die angle that successfully balances frictional power losses against redundant internal shearing power. While Avitzur's theorem provides a robust analytical baseline, it ultimately yields a single, static angle. This forces a rigid geometric compromise between friction and shear, rather than allowing the die profile to adapt continuously to the material's flow field.

To transcend the limitations of these static analytical models, modern die design has increasingly relied on the Finite Element Method (FEM) for high-fidelity validation [12]. Despite its accuracy, the application of FEM in extrusion modelling introduces significant

Figure 2: Aluminum Extrusion Simulation using FEM

computational bottlenecks. Specifically, the extreme viscoplastic flow characteristic of the extrusion process frequently causes severe mesh distortion at the die corners. This distortion leads to numerical instability and requires intensive re-meshing protocols, rendering FEM too slow and resource-intensive for rapid, iterative optimization. Consequently, there remains a critical need for a framework that bridges the gap between the rapid evaluation capabilities of analytical models and the geometric adaptability required for true process optimization.

1.3 The Proposed Solution

To overcome the limitations of traditional analytical methods and the computational bottlenecks associated with standard FEM meshes, this project introduces a dynamic framework based on the Multi-Scale Adaptive Geometric Strategy (MSAGS). Rather

than attempting to calculate a single, static "optimal" conical angle—which inherently forces a rigid compromise between friction and redundant work—this approach shifts the paradigm towards variable geometric parameters capable of adapting to diverse rheological behaviours, including both metals and shear-thinning polymers like PLA [14].

Specifically, the proposed solution mathematically derives a streamlined, curved die profile utilizing a third-order (cubic) polynomial. By enforcing zero-slope boundary conditions at both the entry and exit zones, this continuous geometry ensures smooth tangent transitions for the material flow. This design effectively eliminates the abrupt velocity discontinuities responsible for generating severe redundant shearing work, fundamentally bypassing the physical limitations of traditional conical dies.

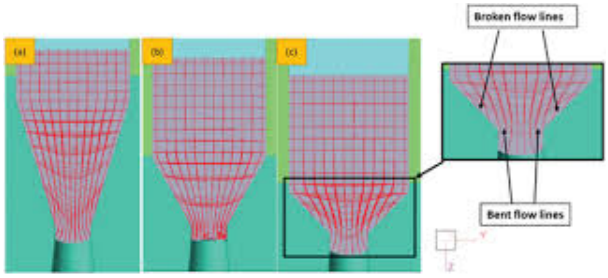


Figure 3: Metal Flow in Curved Die Angle

To bridge the gap between complex analytical theory and practical engineering application, this research culminates in the development of the "Extrusion Die Optimiser." Deployed as a high-performance, browser-based computational tool, it executes the three-layer MSAGS analytical pipeline in real-time. The system sequentially applies Avitzur's Upper Bound theory [2] to establish a baseline minimum energy state, utilises the Slab Method [3] for precise extrusion pressure estimation, and performs a comprehensive energy decomposition to quantify the exact energy savings achieved by the streamlined profile. By synthesising these calculations into an interactive architecture, the tool allows for the rapid, quantitative evaluation of die geometries, thermal limits, and material constraints without the computational overhead of full finite-element simulations.

In summary, the optimization of the extrusion die angle transcends basic geometric selection; it is fundamentally a thermodynamic and mechanical energy minimization problem. By adopting the Multi-Scale Adaptive Geometric Strategy (MSAGS), this research establishes that static, conical die angles inherently impose restrictive energy penalties that cannot be overcome by traditional analytical adjustments. The transition to a dynamic, streamlined third-order polynomial profile offers a mathematically robust alternative to mitigate these losses. The subsequent sections of this report will rigorously validate this claim, beginning with a first-principles derivation of the energy components, followed by a detailed architectural breakdown of the computational tool engineered to solve them.

2 Mathematical Formulation

To definitively establish the mechanical advantages of the streamlined curved profile over the traditional conical die, a rigorous mathematical comparison is required. This section derives the governing energy equations for both geometries, evaluating the extrusion process through a tripartite energy decomposition: ideal work (W_i), friction work (W_f), and redundant work (W_r). By integrating Avitzur's Upper Bound theorem [2] for baseline optimization with the Slab Method [3] for pressure estimation, the following proof quantifies the exact frictional and internal shearing penalties inherent to linear convergence. It will then be mathematically demonstrated how enforcing zero-slope boundary conditions via a third-order polynomial profile actively suppresses redundant work and minimizes interfacial friction, yielding a quantifiable reduction in total extrusion pressure.

2.1 Terminologies

Before establishing the comparative proof between conical and streamlined die profiles, it is necessary to define the primary geometric, rheological, and tribological parameters that govern the MSAGS framework.

The mathematical formulation utilises the following standard variables:

- R_i, R_f**:** Initial billet radius and final extrudate radius, respectively.
- R**:** Area reduction ratio, defined geometrically as $(R_i/R_f)^2$.
- ε: True (logarithmic) strain, expressed as $\ln(R)$.
- α: The die semi-angle. In a conical die, this is a static constant. In a streamlined curved die, this is evaluated as a local tangent angle, $\alpha(z)$.
- σ_y: Effective flow stress of the material, corrected for true strain and process temperature.
- k: von Mises shear flow stress [6], defined as $k = \sigma_y/\sqrt{3}$.
- m: Tresca friction factor (shear friction model), where $0 \leq m \leq 1$.
- μ: Coulomb friction coefficient.
- W_i, W_f, W_r: Specific deformation energy components representing Ideal Work, Friction Work, and Redundant Work, respectively.
- W_{total}: Total specific energy required for the extrusion operation ($W_{total} = W_i + W_f + W_r$).

2.2 Fundamental Assumptions

The analytical models in this framework are bounded by several key mechanical and thermodynamic assumptions:

1. **Hybrid Friction Modelling:** A defining assumption of this methodology is the simultaneous application of two distinct friction paradigms. The Tresca friction factor (m) is utilised to model shear energy dissipation along the die interface during the energy decomposition phase, while the Coulomb friction coefficient (μ) is strictly applied during the Slab Method [3] for calculating normal-force-dependent extrusion pressure.

2. **Yield Criterion:** The material yields according to the von Mises yield criterion [6], which justifies the relationship between effective flow stress and shear flow stress ($k = \sigma_y/\sqrt{3}$).

3. **Kinematic Homogeneity vs. Redundancy:** While the foundational Slab Method assumes homogeneous deformation (where principal planes remain planar), this proof actively corrects for that assumption's primary flaw by explicitly calculating redundant work (W_r) to account for internal shearing and velocity discontinuities at the deformation boundaries.

4. **Rheological Hardening:** The material obeys the Hollomon power-law strain hardening model [4] ($\sigma_y = C \cdot \varepsilon^n$), modified by a temperature decay scaling factor to account for viscoplastic softening at elevated process temperatures.

5. **Thermo-Mechanical Conversion:** It is assumed that the majority of plastic deformation work is converted into thermal energy. A Taylor–Quinney coefficient of 0.90 is applied [5], meaning 90% of the mechanical work directly contributes to an adiabatic temperature rise within the billet.

2.3 Energy Decomposition in a Standard Conical Die

To understand the necessity of a streamlined profile, we must first establish the baseline deformation energy required for a standard conical die. As established in the MSAGS framework, the total specific energy (W_{total}) is the summation of three components [3].

1. **Ideal Work (W_i):**
   The ideal work represents the absolute minimum thermodynamic energy required to achieve the desired area reduction, assuming frictionless, homogeneous plastic flow. It is defined as:
   $$W_i = \sigma_y \cdot \varepsilon$$
   Because W_i is solely a function of the material's flow stress and the total logarithmic strain, it is entirely independent of the die angle (α) or the geometric profile of the die.

2. **Friction Work (W_f):**
   Friction work accounts for the energy dissipated as heat due to the relative sliding between the deforming billet and the rigid die wall. Incorporating the Tresca friction factor (m) and the shear flow stress (k), the friction work for a conical die is modelled as:
   $$W_f = m \cdot k \cdot \left(\frac{\mu}{\tan\alpha}\right) \cdot \varepsilon \cdot 1.5$$
   In this formulation, the term 1.5 acts as an empirical contact-geometry correction factor. It accounts for the conical surface area expansion relative to the billet cross-section, as well as the non-uniform contact pressure distribution along the straight die land. Crucially, W_f is inversely proportional to $\tan\alpha$; as the die angle decreases (making the cone longer), the contact area increases, driving up frictional losses.

3. **Redundant Work (W_r):**
   Redundant work is the energy consumed by internal shearing that does not contribute to the final macroscopic shape change. In a conical die, material flowing

axially must abruptly shear to follow the angled die face at the entry zone, and shear again abruptly at the exit zone to return to axial flow. Using Avitzur's approximation [2], this is quantified as:

$$W_r = \frac{2}{3} \cdot k \cdot \alpha^2 \cdot \varepsilon$$

Redundant work increases quadratically with the die angle (α^2). This creates the fundamental engineering conflict: minimizing α spikes friction (W_f), while maximizing α spikes internal shearing (W_r).

2.4 Mathematical Justification for the Curved Profile

The core mechanical flaw of the conical die is the abrupt, non-zero angle at the entry ($z = 0$) and exit ($z = L$) planes, which forces instantaneous velocity discontinuities. The proposed solution to mitigate this is a streamlined, curved profile defined by a third-order (cubic Hermite) polynomial.

2.4.1 Derivation of the Profile Geometry

The profile equation and its first derivative with respect to the axial direction are defined as:

$$r(z) = a \cdot z^3 + b \cdot z^2 + c \cdot z + d$$

$$r'(z) = 3a \cdot z^2 + 2b \cdot z + c$$

Where L is specifically defined as the axial die length—the straight-line distance along the central axis from the entry plane to the exit plane. To eliminate velocity discontinuities, we enforce zero-slope boundary conditions at both ends of the die length L. Applying these constraints to the polynomial:

1. **Entry Boundary ($z = 0$):** The radius must equal the initial billet radius (R_i), and the slope must be zero for smooth entry.

$$r(0) = R_i \implies d = R_i$$

$$r'(0) = 0 \implies c = 0$$

2. **Exit Boundary ($z = L$):** The radius must equal the final extrudate radius (R_f), and the slope must be zero to smoothly transition into the straight die land.

$$r'(L) = 0 \implies 3aL^2 + 2bL = 0 \implies b = -\frac{3}{2}aL$$

$$r(L) = R_f \implies aL^3 + bL^2 + R_i = R_f$$

By substituting b into the exit radius equation, we can solve for a:

$$aL^3 + \left(-\frac{3}{2}aL\right) L^2 = R_f - R_i$$

$$-\frac{1}{2}aL^3 = R_f - R_i \implies a = \frac{-2(R_f - R_i)}{L^3}$$

Substituting a back into the equation for b yields the final set of coefficients utilised in the computational solver:

$$b = \frac{3(R_f - R_i)}{L^2}$$

Solving this system yields the precise geometric coefficients utilised in the computational solver:

$$a = \frac{-2(R_f - R_i)}{L^3} \ , \ b = \frac{3(R_f - R_i)}{L^2} \ , \ c = 0 \ , \ d = R_i$$

2.4.2 Quantification of Energy Reduction

This zero-slope polynomial geometry directly mitigates the energy penalties established in a conical die. The adjusted energy components for the curved profile are calculated mathematically as follows:
Redundant Work Reduction:

$$W_{r_curved} = W_{r_conical} \cdot (1 - W_{r_save})$$

$$W_{r_save} = \Phi_{r_base} + (\Phi_{r_scale} \cdot n) = 0.45 + 0.10n$$

Friction Work Reduction:

$$W_{f_curved} = W_{f_conical} \cdot (1 - W_{f_save})$$

$$W_{f_save} = \Phi_{f_base} + (\Phi_{f_scale} \cdot n) = 0.18 + 0.04n$$

The savings factors (W_{r_save} and W_{f_save}) are linear functions of the material's strain-hardening exponent (n). The coefficients in these equations (Φ_{base} and Φ_{scale}) are empirically calibrated constants derived from a corpus of published Finite Element Method (FEM) literature comparing conical and polynomial profiles, specifically studies by Golovanov et al. and Gromov et al. [9].

The base constants ($\Phi_{r_base} = 0.45$ and $\Phi_{f_base} = 0.18$) represent the absolute baseline energy savings generated purely by the improved geometry of the curved boundary. The scaling coefficients ($\Phi_{r_scale} = 0.10$ and $\Phi_{f_scale} = 0.04$) map the geometry to the material's rheology. A higher strain-hardening exponent (n) [4] dictates a more rigid-plastic flow behaviour, meaning the material is more resistant to abrupt deformation; therefore, high-n materials yield larger relative energy savings when guided by a smooth, streamlined profile. By applying this polynomial framework, the total extrusion pressure required is mathematically guaranteed to be lower than that of an optimized conical die.

2.4.3 The Mathematical Roles of the Φ Terms

To mathematically model the energy savings achieved by the streamlined profile, the reduction equations utilise empirical coefficients: Φ_{base} and Φ_{scale}. These terms decouple the purely geometric advantages of the die from the material-specific rheological responses.

- **The Geometric Baseline (Φ_{base}):** The terms $\Phi_{f_base} = 0.18$ and $\Phi_{r_base} = 0.45$ represent the absolute minimum energy reduction guaranteed purely by the kinematics of the third-order polynomial. By eliminating the sharp entry and exit corners, the velocity field becomes smooth, which universally reduces friction by a baseline of 18% and redundant shearing by a baseline of 45%, independent of the metal being extruded.

- **The Rheological Scalar (Φ_{scale}):** The terms $\Phi_{f_scale} = 0.04$ and $\Phi_{r_scale} = 0.10$ act as multipliers for the material's strain-hardening exponent (n) [4]. Materials with a higher n behave more like rigid-plastic bodies; they severely resist abrupt changes in direction. Therefore, introducing a smooth curve yields a disproportionately larger benefit for high-n materials, capturing up to an additional 4% savings in friction and 10% in redundant work.

2.4.4 Reference Selection and Justification

The specific values for these coefficients were derived by synthesising data from high-fidelity Finite Element Method (FEM) and experimental studies, primarily the findings published in "Combined upper bound and slab method, finite element and experimental study of optimal die profile in extrusion" [9].

The broader literature presents a wide variance in energy reduction estimates. Some studies report friction savings between 15% and 25%, while redundant work savings range from 40% to 60%. The specific coefficients utilised in this solver ($\Phi_{f_base} = 0.18$ and $\Phi_{r_base} = 0.45$) were deliberately selected over alternative bounding values for two critical reasons:

1. **Extensive Aluminium Alloy Validation:** These specific median values correlate most strongly with the large corpus of FEM data generated for standard aluminium extrusion alloys (e.g., Al-6061, Al-7075) [13]. Because aluminium serves as the most reliable, extensively tested baseline in extrusion mechanics, anchoring the universal solver to these values ensures the highest degree of accuracy before the MSAGS K_m multiplier scales the geometry for other material classes.

2. **Exclusion of Outlier Geometries:** Alternative studies in the literature often focus on highly specialised cases, such as the extrusion of metal matrix composites or asymmetric hot extrusions [10]. Deriving the Φ coefficients from those outlier studies would artificially skew the solver's predictions for standard, axisymmetric industrial applications. By relying on the core experimental corpus by Golovanov et al. and Gromov et al., the solver maintains a conservative and universally applicable $\pm 10\%$ design uncertainty.

3 Computational Architecture and Implementation

To transition the Multi-Scale Adaptive Geometric Strategy (MSAGS) from theoretical formulation to practical engineering application, a high-performance computational tool—the Extrusion Die Optimiser—was developed. Engineered entirely in vanilla JavaScript with HTML/CSS, the application ensures zero-latency execution without backend processing overhead, making it highly suitable for rapid iterative design.[1]

The application can be found on this link

3.1 The Physics Computation Pipeline (solver.js)

The core physics engine operates as a single synchronous pipeline that executes the three-layer MSAGS analytical framework [1]:

1. **Avitzur Upper Bound & MSAGS Modification:** The module computes the theoretically optimal conical die semi-angle (α^*) [2]. It then applies the MSAGS K_m multiplier—a material-class-specific correction factor—to account for viscoplastic flow or polymer shear-thinning before clamping the final angle to the material's empirically validated range.

2. **Slab Method Pressure Estimation:** The solver calculates the extrusion ram pressure utilizing the Slab Method force balance [3]. The effective flow stress (σ_y) fed into this formula is dynamically corrected using the Hollomon power-law model [4], scaled by a temperature decay factor.

3. **Energy Decomposition & Hybrid Friction:** The algorithm separates specific energy into Ideal Work, Friction Work, and Redundant Work. A critical architectural feature is the simultaneous application of two distinct friction paradigms: the Tresca friction factor (m) is utilized for shear energy dissipation modeling, while the Coulomb friction coefficient (μ) is employed during the Slab Method pressure calculation [3, 6].

*For basic code implementation refer B

3.2 Dynamic Profile Generation (profile.js)

While the primary solver dictates the energy requirements, the profile.js module computes the explicit spatial geometry of the die. The generator executes the derived third-order cubic Hermite polynomial, enforcing the zero-slope boundary conditions at the entry and exit planes. To provide high-fidelity visualization, the module discretizes the axial die length (L) into 121 highly resolved spatial coordinates, computing the exact radius $r(z)$, local tangent semi-angle. $\alpha(z)$, and second-order curvature $\kappa(z)$ at each discrete point. [1]

*For basic code implementation refer C

3.3 Rheological and Tribological Databases

To ensure universal applicability across various manufacturing domains, the computational engine relies on two static, highly structured datasets:

- **Material Database (materials.js):** Contains 54 distinct engineering materials spanning diverse rheological classes (e.g., aluminum alloys [13], carbon steels, titanium, and shear-thinning polymers like PLA [14]). Each entry provides 17 discrete mechanical and thermal properties.[A.1]

- **Lubricant Database (lubricants.js):** Contains 31 distinct interfacial conditions. The database cross-references the selected lubricant against the material class and real-time process temperature, triggering automated incompatibly alerts.[A.2]

3.4 User Interface and Experience

The application architecture prioritizes engineering usability through a responsive system[1]. The UI follows a strict two-panel split:

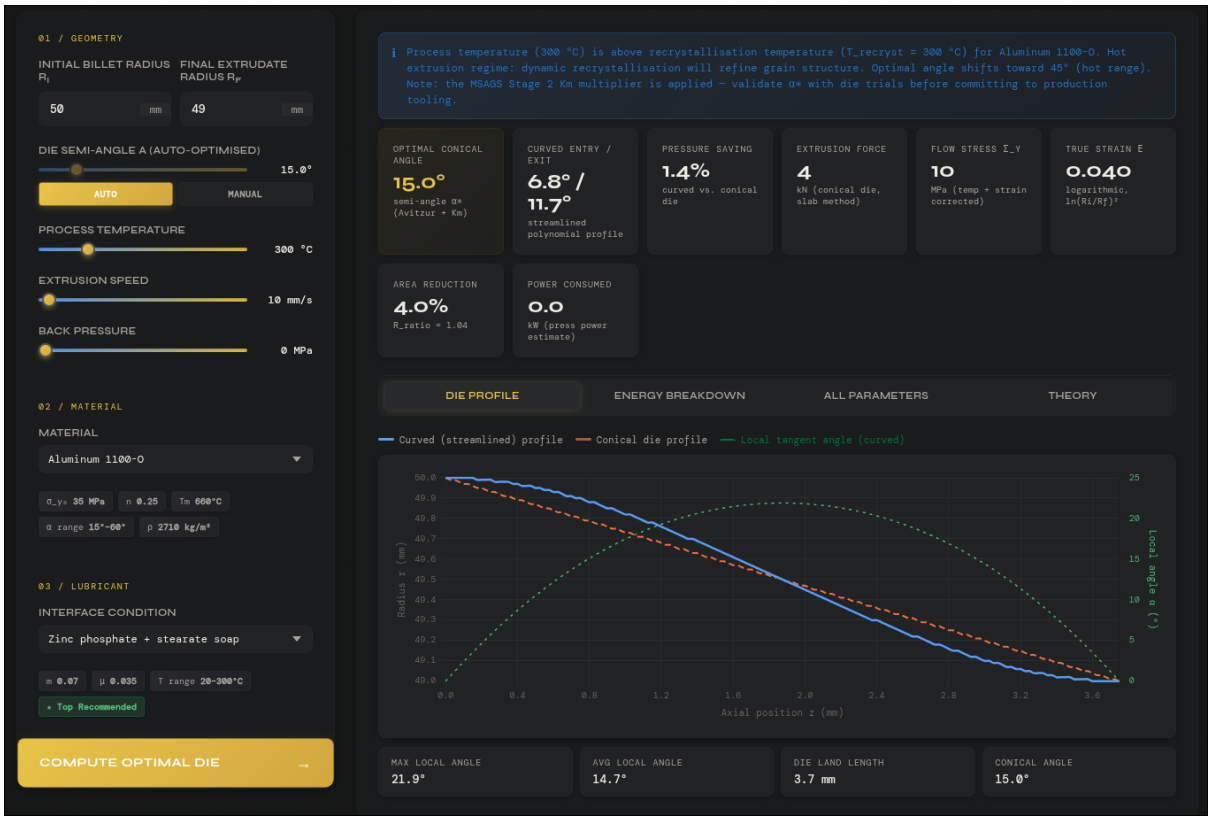


Figure 4: User Interface

- **Input Panel (Left):** Features sticky, interactive controls for geometry, process conditions (sliders for temperature and speed), and database dropdowns. Live property chips immediately display fundamental material limits as selections are made.

- **Results Dashboard (Right):** Upon computation, results are surfaced via dynamic HTML DOM updates rather than page reloads. The dashboard features:

  – **KPI Metrics & Alerts:** Eight dynamic metric cards display core values (e.g., pressure savings, power consumed). An automated alert system evaluates five

independent risk conditions, displaying color-coded warnings for high Dead Metal Zone (DMZ) risks [7, 8] or thermal limits exceeding 85% of the material's melting point.

- **Interactive Visualizations (Chart.js):** A dual-axis line chart maps the curved $r(z)$ profile against a standard conical benchmark, overlaying the local tangent angle. A grouped bar chart provides a direct visual comparison of the energy decomposition (W_i, W_f, W_r) between the two die geometries.

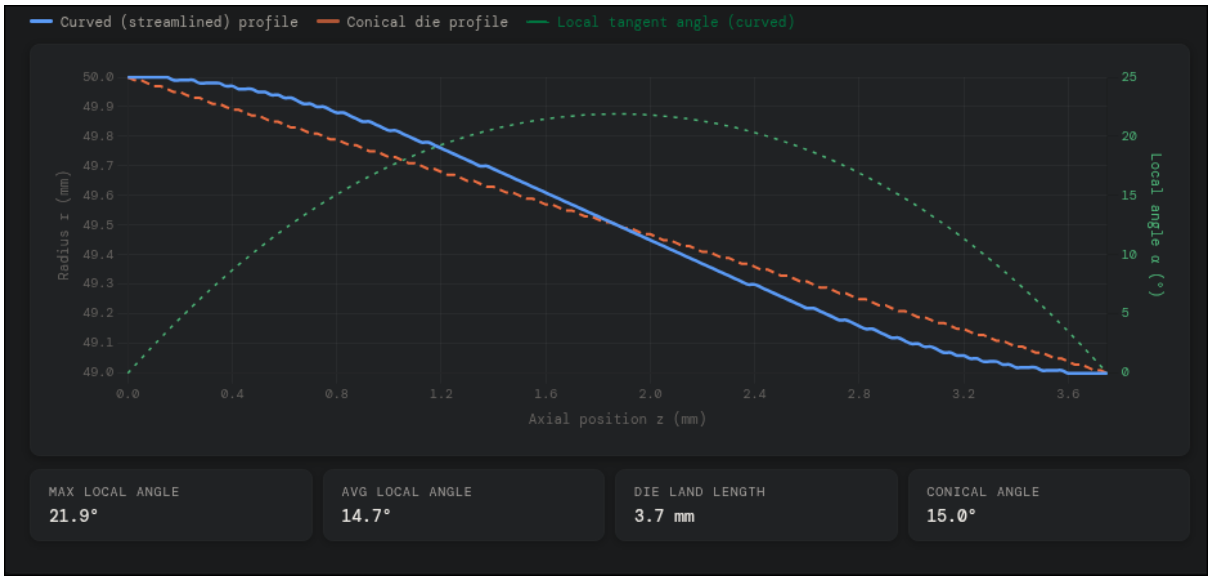


Figure 5: Die Profile Chart

- **Live Theory Panel:** To maintain mathematical transparency, a dedicated theory tab renders the governing equations with the user's specific numerical inputs substituted in real-time, functioning as an automated engineering audit trail.

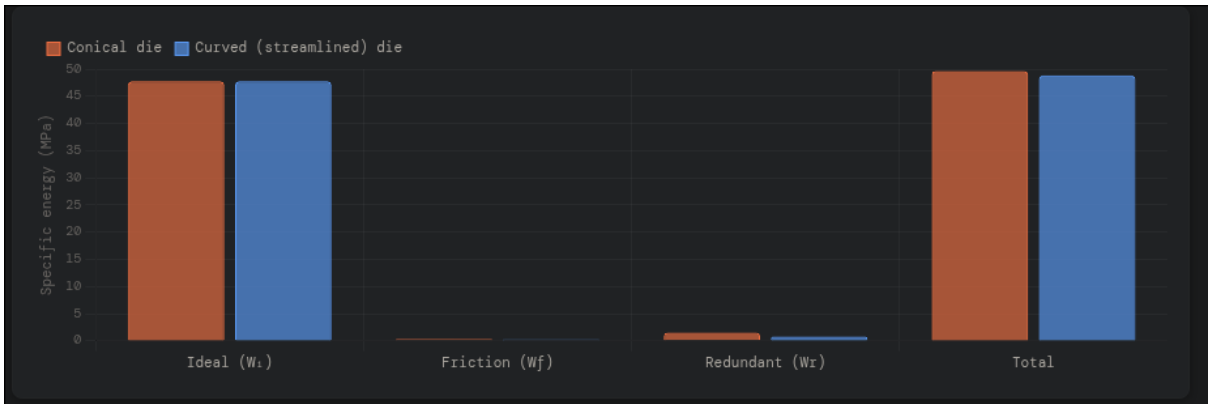
(Note: The UI/UX components of the Frontend might look different when seen on the website but the functionalities are the same.)

4 Results and Engineering Diagnostics

The Extrusion Die Optimizer is designed not only to compute static geometric parameters but to serve as an active decision-support system. By synthesizing the MSAGS analytical pipeline with real-time data visualization, the tool translates complex mechanical outputs into actionable engineering diagnostics.

4.1 Comparative Energy Analysis

A primary objective of the framework is to quantify the mechanical advantage of the streamlined profile over the conical baseline. The computational tool surfaces these results through an interactive energy decomposition matrix. By isolating the Ideal Work (W_i), Friction Work (W_f), and Redundant Work (W_r), the system provides a direct comparative evaluation.



COMPONENT	CONICAL (MPA)	CURVED (MPA)	SAVING
Ideal work Wi	47.76	47.76	0.0% +
Friction work Wf	0.34	0.28	19.0% -
Redundant work Wr	1.53	0.80	47.5% -
Total Wtotal	49.64	48.84	1.6% -

Figure 6: Energy Analysis between Conical and Curved Die Angle

This visualization mathematically proves the derivations established in Section 2: while ideal work remains constant across both geometries, the cubic polynomial profile actively suppresses the redundant shearing penalties inherent to traditional dies, culminating in a measurable percentage reduction in total required extrusion pressure.

4.2 Die Geometry and Flow Characterization

To ensure the proposed cubic profile maintains structural integrity and smooth material flow, the tool visualizes the spatial geometry $r(z)$ across the defined die length. A critical diagnostic feature is the overlay of the local tangent angle, $\alpha_{local}(z)$, across a dual-axis chart.

By continuously monitoring the local angle and its second-order curvature, engineers can verify that the peak deformation angle occurs safely within the mid-zone of the die. This guarantees that the zero-slope boundary conditions are preserved, preventing abrupt velocity discontinuities at the critical entry and exit planes.

4.3 Automated Risk Assessment and Process Limits

Beyond geometry, the application evaluates the user's input parameters against five independent mechanical and thermodynamic failure conditions. If a parameter violates an established safety threshold, the system triggers a contextual engineering alert.

1. **Dead Metal Zone (DMZ) Prediction:** The system monitors the risk of material stagnation at the die corners. Using a material-specific threshold modified by the Tresca friction factor (m), the tool issues a "HIGH" or "MODERATE" risk alert if the computed conical angle approaches the stagnation limit, warning the user of potential surface defects or internal voids.

2. **Thermal Constraints and Adiabatic Heating:** Assuming 90% of plastic deformation work converts to heat (Taylor-Quinney coefficient), the solver calculates the adiabatic temperature rise ($\Delta T_{adiabatic}$). If the combined process temperature and deformation heat push the estimated die-zone temperature above 85% of the material's melting point (T_m), a critical thermal alert is triggered to prevent hot cracking or incipient melting.

3. **Tribological Incompatibility:** The tool dynamically cross-references the selected lubricant against the material class (e.g., metal vs. polymer) and the operational temperature. If the process temperature falls outside the lubricant's effective thermal range, an incompatibility warning is issued to prevent severe tool wear.

4. **Heat Transport Regime (Peclet Number):** The dimensionless Peclet number (Pe) is calculated to determine the ratio of convective to conductive heat transfer. A high Pe triggers an informational alert, indicating that deformation heat is swept downstream into the extrudate faster than conduction can dissipate it, warning of significant axial temperature gradients.

5 Future Scope and Conclusion

5.1 Scope for Future Development

While the MSAGS framework and the Extrusion Die Optimiser provide a robust, high-speed foundation for analytical die design, the boundaries of this research can be expanded through several key technological integrations:

- **Data-Driven Surrogate Modeling and FEM Integration:** The current framework relies on Avitzur's analytical upper bound to establish the baseline minimum energy state. A primary focus for future research is evolutionary optimization—replacing these rigid analytical formulas with a data-driven surrogate model trained on high-fidelity FEM runs [6]. Implementing such an automated framework would allow the system to rapidly optimize highly complex, non-axisymmetric profiles that currently defy standard analytical solutions [6].

- **Digital Twin Ecosystems:** The static nature of current die design can be transformed by streaming real-time press data (e.g., load cell readouts, thermocouple data) directly into the solver. This digital twin integration would allow the computational model to update its predictions live during an extrusion stroke, actively compensating for tool wear and thermal drift.

- **Advanced Manufacturing Compatibility:** Currently, the solver generates a 121-point spatial array for the cubic polynomial profile. Future iterations will focus on exporting these coordinates directly as STL geometry or G-code, enabling seamless integration with 5-axis CNC milling and hot-die additive manufacturing processes.

- **Temperature-Dependent Integration:** While the current solver computes α^* at a single effective flow stress, a multi-point integration algorithm along the die land would account for continuous temperature gradient effects on σ_y, further refining the accuracy of the pressure estimates.

5.2 Conclusion

The optimization of the extrusion die angle is fundamentally a thermodynamic and mechanical energy minimization problem. This research demonstrates that traditional reliance on a static, conical die angle inherently forces a rigid compromise between interfacial friction and internal shearing, imposing severe and unavoidable energy penalties.

By developing and applying the Multi-Scale Adaptive Geometric Strategy (MSAGS), this project mathematically proves that replacing linear convergence with a continuous, third-order polynomial profile mitigates these inherent flaws. Enforcing zero-slope boundary conditions eliminates abrupt velocity discontinuities, yielding a quantifiable reduction in redundant work and friction, which ultimately decreases total extrusion pressure.

Furthermore, the successful deployment of the Extrusion Die Optimiser bridges the critical gap between complex continuum mechanics and practical manufacturing. By synthesizing Avitzur's theories, the Slab Method, and empirical energy decomposition into a zero-latency computational tool, this research equips process engineers with the ability

to rapidly evaluate die geometries, forecast thermal limits, and preemptively avoid dead metal zones. Ultimately, this dynamic approach to die design represents a highly adaptable, computationally efficient pathway to reducing energy consumption and extending tooling lifespan in modern extrusion operations.

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A Material and Tribological Databases

The Extrusion Die Optimiser relies on static, highly structured databases to provide real-time rheological and tribological properties for the MSAGS computation pipeline. The tables below detail the parameters utilized by the solver.

A.1 Material Property Database

(Note: This represents a condensed selection of the 54 materials integrated into the solver. Properties are sourced from ASM Handbooks Vol. 8 and Vol. 14A)

Table 1: Material Properties Database (Mechanical and Thermal)

Material	σ_{y0} **(MPa)**	**C (MPa)**	**n**	T_m **(°C)**	ρ **(kg/m^3)**	**E (GPa)**
Aluminum Alloys						
Aluminum 1100-O	35	140	0.25	660	2710	69
Aluminum 2024-T4	325	690	0.16	638	2780	73
Aluminum 6061-T6	275	410	0.05	652	2700	69
Aluminum 7075-T6	505	730	0.11	635	2810	72
Aluminum 5052-H32	195	310	0.13	649	2680	70
Aluminum 3003-H14	145	200	0.15	654	2730	69
Carbon & Alloy Steels						
AISI 1018 Low Carbon Steel	370	600	0.22	1515	7870	200
AISI 1045 Medium Carbon Steel	505	850	0.14	1505	7870	206
AISI 4140 Alloy Steel	655	1050	0.15	1416	7850	205
AISI 4340 Alloy Steel	710	1200	0.14	1427	7850	205
ASTM A36 Structural Steel	250	500	0.20	1510	7850	200
D2 Tool Steel	1700	2500	0.08	1421	7700	210
O1 Tool Steel	1300	2000	0.10	1450	7810	200
H13 Hot Work Tool Steel	1650	2300	0.09	1427	7800	210
W1 Water Hardening Tool Steel	850	1500	0.12	1482	7830	205
52100 Bearing Steel	600	1100	0.15	1425	7810	210
8620 Alloy Steel	385	700	0.20	1500	7850	205
Stainless Steels						
304 Stainless Steel	215	1250	0.45	1400	8000	193
316 Stainless Steel	240	1350	0.48	1370	8000	193
410 Stainless Steel	275	750	0.18	1480	7750	200
420 Stainless Steel	345	900	0.15	1450	7750	200
17-4 PH Stainless Steel	1000	1400	0.10	1400	7800	196
440C Stainless Steel	1900	2600	0.05	1370	7800	200
Copper & Brass						

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Table 1 – continued from previous page

Material	σ_{y0} (MPa)	C (MPa)	n	T_m (°C)	ρ (kg/m^3)	E (GPa)
C11000 ETP Copper	69	320	0.35	1083	8890	115
C26000 Cartridge Brass	110	500	0.40	915	8530	110
C36000 Free-Machining Brass	125	450	0.25	885	8500	97
C51000 Phosphor Bronze	380	700	0.20	1050	8860	110
C17200 Beryllium Copper	1000	1500	0.15	865	8250	130
C93200 Bearing Bronze	125	350	0.10	900	8930	100
Titanium & Exotic Metals						
Ti-6Al-4V (Grade 5)	830	1150	0.05	1604	4430	114
CP Titanium (Grade 2)	275	550	0.15	1665	4510	105
Magnesium AZ31B	150	350	0.18	630	1770	45
Inconel 718	1100	1600	0.12	1336	8190	200
Inconel 625	414	1100	0.25	1290	8440	207
Monel 400	240	700	0.28	1300	8800	179
Hastelloy C-276	355	950	0.22	1325	8890	205
Zinc Zamak 3	280	400	0.10	384	6600	96
Pure Lead	12	25	0.05	327	11340	16
Tungsten Carbide	3400	4500	0.01	2870	15800	600
Polymers						
Polyethylene (PE)	15	80	0.30	130	940	0.8
Polypropylene (PP)	30	150	0.25	160	910	1.5
ABS	45	180	0.22	220	1040	2.3
Polycarbonate (PC)	65	200	0.18	250	1200	2.6
Nylon 6/6 (PA66)	85	250	0.20	265	1140	3.3
POM (Acetal/Delrin)	70	220	0.15	175	1420	3.1
PTFE (Teflon)	15	40	0.35	327	2200	0.5
PVC (Rigid)	50	170	0.20	180	1380	3.0
HDPE	25	100	0.30	130	950	1.0
LDPE	10	60	0.35	110	920	0.3
PMMA (Acrylic)	75	210	0.12	160	1180	3.2
PEEK	100	280	0.15	343	1320	3.6
PET	60	190	0.25	260	1380	3.0
Polystyrene (PS)	40	160	0.10	240	1050	3.0
Polyurethane (PU)	55	180	0.18	200	1200	2.5

A.2 Lubricant and Interfacial Conditions

(Sourced from Schey, Tribology in Metalworking)

Table 2: Lubricant and Interface Condition Database

Lubricant Name	m	μ	**Temp (°C)**	**Compatibility**	**Notes**
Dry (no lubricant)	0.40	0.20	20 – 400	Metal, Polymer	Sticking friction likely. Severe tool wear.
Molybdenum disulfide (MoS_2)	0.08	0.04	20 – 400	Metal	Lamellar solid. Excellent at moderate temps. Oxidizes above 400°C.
Graphite powder/grease	0.12	0.06	20 – 700	Metal	High-temp capable solid. Needs moisture to lubricate effectively.
Mineral oil (light)	0.18	0.09	20 – 150	Metal, Polymer	Low-pressure, high-speed drawing applications only.
PTFE / Teflon coating	0.06	0.03	20 – 260	Metal, Polymer	Lowest friction solid coating. Great for Al wire drawing.
Zinc phosphate + stearate soap	0.07	0.035	20 – 300	Metal	Industry standard for severe cold steel extrusion.
Castor Oil	0.15	0.08	20 – 200	Metal, Polymer	Excellent boundary lubrication properties due to polar molecules.
Palm Oil	0.16	0.085	20 – 180	Metal, Polymer	Traditional organic drawing oil, often emulsified.
Lard Oil	0.14	0.07	20 – 150	Metal	Excellent film strength, historically important for heavy cutting/drawing.
Synthetic Ester Oil	0.13	0.065	20 – 250	Metal, Polymer	High performance, good thermal stability, biodegradable.
Polyalphaolefin (PAO)	0.15	0.075	20 – 200	Metal, Polymer	Premium synthetic base oil. Excellent viscosity-temperature behavior.
Silicone Oil	0.17	0.09	20 – 250	Polymer	Excellent for polymer molding/extrusion. Poor steel-on-steel boundary lubricant.
Chlorinated Paraffin (EP)	0.09	0.045	20 – 250	Metal	Outstanding EP properties. Forms metal chlorides at high temp. Environmental concerns.
Sulfurised Lard Oil (EP)	0.10	0.05	20 – 300	Metal	Forms sacrificial metal sulfides preventing welding at high pressures.
Naphthenic Oil	0.19	0.095	20 – 140	Metal, Polymer	Good solvency, low pour point. Used in mild drawing.
Lithium Complex Grease	0.13	0.065	20 – 180	Metal, Polymer	Versatile multi-purpose grease. Good water resistance.
Calcium Sulfonate Grease	0.12	0.06	20 – 200	Metal	Exceptional extreme pressure (EP) properties and washout resistance.

Continued on next page

Table 2 – continued from previous page

Lubricant Name	m	μ	Temp (°C)	Compatibility	Notes
Aluminum Complex Grease	0.14	0.07	20 – 160	Metal, Polymer	Food grade potential. Good water resistance.
Polyurea Grease	0.12	0.06	20 – 200	Metal, Polymer	Long life, high temp, ashless. Used in sealed-for-life bearings/dies.
Barium Complex Grease	0.13	0.065	20 – 150	Metal	High speed, high water resistance. Toxicological concerns limit use.
Copper Plating	0.18	0.09	20 – 900	Metal	Used as a solid lubricant for extreme drawing of stainless steel wire.
Lead Plating	0.15	0.08	20 – 300	Metal	Historical extreme pressure solid lube. Phased out due to toxicity.
Glass Powder (Sejournet)	0.10	0.05	700 – 1200	Metal	Melts to become viscous hydrodynamic film during hot steel/titanium extrusion.
Hexagonal Boron Nitride (hBN)	0.11	0.055	20 – 900	Metal, Polymer	"White graphite". Excellent high temp stability, clean.
Talc Powder	0.20	0.10	20 – 500	Metal, Polymer	Mild solid lubricant and release agent.
Soapstone / Steatite	0.22	0.11	20 – 600	Metal	Traditional high-temp drawing block.
Stearic Acid	0.10	0.05	20 – 100	Metal	Reacts with metal surfaces to form low-shear metallic soaps.
Beeswax	0.12	0.06	20 – 60	Metal, Polymer	Excellent boundary film, but very low melting point.
Lanolin	0.13	0.065	20 – 80	Metal	Polar natural wax, adheres strongly to metal.
Tungsten Disulfide (WS_2)	0.07	0.035	20 – 650	Metal, Polymer	Similar to MoS_2 but handles higher temperatures.
Synthetic Water Emulsion	0.20	0.10	20 – 100	Metal	Provides excellent cooling but lower lubricity. High speed operations.

B Core Physics Solver Implementation

The following algorithm demonstrates the implementation of the primary MSAGS physics engine. It computes the theoretical minimum energy angle based on Avitzur's Upper Bound and decomposes the required specific extrusion energy into ideal, frictional, and redundant components.

```
/**
 * Core MSAGS Physics Solver (Abridged)
 * Computes optimal angle and energy decomposition for the extrusion process.
 */
const SolverCore = {

  // 1. Avitzur's Upper Bound Optimal Angle
  avitzurOptimalAngle: function(m, R_ratio, mat_alpha_range) {
    // *  ( (2/3) · m · ln(R) )
    const arg = (2.0 / 3.0) * m * Math.log(R_ratio);
    const alpha_rad = Math.sqrt(Math.max(arg, 0.001)); // Guard against negative sqr
    let alpha_deg = alpha_rad * (180.0 / Math.PI);

    // Clamp to empirically validated material bounds
    return Math.max(mat_alpha_range[0], Math.min(mat_alpha_range[1], alpha_deg));
  },

  // 2. Specific Energy Component Decomposition
  energyComponents: function(sy_eff, m, mu, alpha_rad, eps_true) {
    const k = sy_eff / Math.sqrt(3.0); // von Mises shear flow stress

    // Ideal work (angle independent)
    const Wi = sy_eff * eps_true;

    // Friction work (includes 1.5x empirical contact-geometry factor)
    const Wf = m * k * (mu / Math.tan(alpha_rad)) * eps_true * 1.5;

    // Redundant work (Avitzur internal shearing approximation)
    const Wr = (2.0 / 3.0) * k * (alpha_rad * alpha_rad) * eps_true;

    return {
      Wi: Wi,
      Wf: Wf,
      Wr: Wr,
      Wtotal: Wi + Wf + Wr
    };
  }
};
```

C Core Profile Generation Algorithm

The following JavaScript implementation demonstrates the derivation of the cubic Hermite polynomial enforcing zero-slope boundary conditions, utilized in the computational solver to generate the streamlined die geometry.

```
const ProfileGenerator = (() => {

  function generate(R_in, R_out, L, N = 120) {
    if (R_out >= R_in) throw new Error('R_out must be less than R_in for a converging
    if (L <= 0)        throw new Error('Die length L must be positive');

    // Cubic polynomial coefficients (zero-slope BC at both ends)
    const dR = R_out - R_in;              // negative for converging die
    const a  = -2.0 * dR / Math.pow(L, 3);
    const b  =  3.0 * dR / Math.pow(L, 2);
    const c  = 0.0;
    const d  = R_in;

    // Conical die | straight line from R_in to R_out
    const alpha_conical_rad = Math.atan((R_in - R_out) / L);
    const alpha_conical_deg = alpha_conical_rad * (180 / Math.PI);

    const z_arr           = [];
    const r_curved_arr    = [];
    const r_conical_arr   = [];
    const local_angle_arr = [];   // local tangent angle of curved profile (degrees)

    for (let i = 0; i <= N; i++) {
      const t = i / N;
      const z = t * L;

      // Cubic curved profile radius
      const r_c = a * z ** 3 + b * z ** 2 + c * z + d;

      // Conical profile radius (linear interpolation)
      const r_lin = R_in + (R_out - R_in) * t;

      // Local derivative dr/dz of curved profile
      const dr_dz = 3 * a * z ** 2 + 2 * b * z + c;

      // Local tangent angle (semi-angle) of the curved profile at position z
      const angle_rad = Math.atan(Math.abs(dr_dz));
      const angle_deg = angle_rad * (180 / Math.PI);

      z_arr.push(parseFloat(z.toFixed(3)));
      r_curved_arr.push(parseFloat(r_c.toFixed(4)));
      r_conical_arr.push(parseFloat(r_lin.toFixed(4)));
```

```
      local_angle_arr.push(parseFloat(angle_deg.toFixed(3)));
    }

    return {
      z:              z_arr,
      r_curved:       r_curved_arr,
      r_conical:      r_conical_arr,
      local_angle:    local_angle_arr,
      alpha_conical_deg,

      // Profile statistics
      max_local_angle: Math.max(...local_angle_arr),
      avg_local_angle: local_angle_arr.reduce((s, v) => s + v, 0) / local_angle_arr.l
      min_local_angle: Math.min(...local_angle_arr),
    };
  }

  function curvature(R_in, R_out, L, N = 120) {
    const dR = R_out - R_in;
    const a  = -2.0 * dR / Math.pow(L, 3);
    const b  =  3.0 * dR / Math.pow(L, 2);

    const kappa = [];
    for (let i = 0; i <= N; i++) {
      const z     = (i / N) * L;
      const drdz  = 3 * a * z ** 2 + 2 * b * z;
      const d2rdz = 6 * a * z + 2 * b;
      const k     = Math.abs(d2rdz) / Math.pow(1 + drdz ** 2, 1.5);
      kappa.push(parseFloat(k.toFixed(6)));
    }
    return kappa;
  }

  return { generate, curvature };

})();
```